Blind Super-Resolution of Single Remotely Sensed Hyperspectral Image

Zhiyuan Liang, Shuai Wang, Tao Zhang, *Student Member, IEEE*, and Ying Fu, *Senior Member, IEEE*

***Abstract*— Hyperspectral image (HSI) super-resolution has recently advanced with significant progress by utilizing the powerful representation capabilities of deep neural networks (DNNs). These approaches, however, inevitably rely on a sizable amount of training data which can be difficult to acquire for remotely sensed HSIs. In many cases, these methods are designed and tailored for only one or a few specific super-resolution scenarios, making them inflexible for handling images with different unknown degradations. In this article, we introduce a two-step framework for blind remotely sensed HSI super-resolution, where the degradation is unknown. Specifically, in the first step, we propose to leverage the abundant remotely sensed color images to address the data insufficiency for remotely sensed HSI super-resolution. It is achieved by exploring the spatial knowledge from remotely sensed color images with a super-resolution network for a predefined degradation, which is then transferred to HSIs via band-by-band super-resolution. Direct use of the results from the transferred super-resolution network is suboptimal as it neglects the spectral correlations of different bands and the gap between predefined degradation and the real one. To make further refinements, we present an unsupervised scheme that simultaneously refines the super-resolved HSI and the unknown degradation by a nonnegative matrix factorization network and a learnable degradation prior. To validate the effectiveness of our method, we conducted extensive experiments on a variety of remotely sensed HSI datasets. The results demonstrate that our method could generalize on various unknown degradations with superior performance against the state-of-the-art methods.**

***Index Terms*— Blind super-resolution, hyperspectral image (HSI), transfer learning, unsupervised learning.**

I. Introduction

REMOTELY sensed hyperspectral image (HSI) has great potential, because of its abundant information in spectral dimension, such as classification [1], [2], object detecting [3], [4], agriculture [5], [6], geological exploration [7], [8], and more. However, the current mainstream hyperspectral imaging systems [9], [10] usually sacrifice the spatial resolution to guarantee the spectral resolution. It promotes the development of HSI super-resolution techniques, especially single HSI super-resolution that only requires low-resolution (LR) HSI.

The traditional single HSI super-resolution methods, e.g., dictionary learning [11], [12] and low-rank approximation [13], usually rely on the handcraft priors to capture the characteristics of HSIs. However, they are limited to the general patterns of images and could fail to recover the fine-grained details properly. With the rapid development of deep-learning, recent methods [14], [15] of super-resolution for remotely sensed HSIs tend to utilize the deep neural network (DNN) to learn the super-resolution process in a purely data-driven approach. These methods get rid of the handcraft priors and usually achieve much better results. However, their performance is strongly correlated with the quality and quantity of the training data, which is difficultly guaranteed for remotely sensed HSIs. For example, existing methods are usually trained and tested on different regions from a single remotely sensed HSI. This is generally far from being sufficient for training a modern DNN and might lead to severe overfitting.

Recent achievements [14], [15], [16], [17], [18], [19], [20] in single HSI super-resolution are mostly credited to DNNs based on supervised learning. These DNN-based methods are often trained in a supervised manner with pairs of LR and high-resolution (HR) HSIs. Typically, the LR HSI is synthesized from the corresponding HR HSI using a specific degradation model. Once the degradation changes, the supervised DNN-based method needs to be retrained to fit the new degradation [21], [22]. Moreover, the real degradation may be unknown in practice. To address the limitation of the supervised DNN-based method, blind HSI super-resolution methods [23], [24], [25] are proposed to deal with these scenarios where the degradation is unknown. These approaches are usually unsupervised but often require an additional HR reference image that is aligned with the LR HSI. The HR reference image can be a multispectral image or a color image. In a sense, the additional HR image provides spatial meaningful information while the LR HSI offers accurate spectral details. However, the HR reference image with accurate alignment is difficult to obtain in practice, which limits the applications of these unsupervised blind HSI super-resolution methods. Indeed, there are some nonblind super-resolution methods based on spectral unmixing [26], [27], [28] that are proposed to remove the constraint of auxiliary HR reference images. However, these methods focus on recovering HR HSIs using

Manuscript received 20 April 2023; revised 7 June 2023; accepted 20 July 2023. Date of publication 25 August 2023; date of current version 21 September 2023. This work was supported by the National Key R&D Program of China (2022YFC3300704), the National Natural Science Foundation of China (62272022, 62331006 and 62171038), and the Fundamental Research Funds for the Central Universities. *(Zhiyuan Liang and Shuai Wang contributed equally to this work.) (Corresponding author: Ying Fu.)*

Zhiyuan Liang and Tao Zhang are with the School of Computer Science and Technology, Beijing Institute of Technology, Beijing 100081, China (e-mail: zhiyuan_liang@bit.edu.cn; tzhang@bit.edu.cn).

Shuai Wang is with the School of Computer Science and Engineering, Beihang University, Beijing 100191, China (e-mail: wangshuai@buaa.edu.cn).

Ying Fu is with the School of Computer Science and Technology and the MIIT Key Laboratory of Complex-Field Intelligent Sensing, Beijing Institute of Technology, Beijing 100081, China (e-mail: fuying@bit.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3302128

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

traditional algorithms, without designing specialized modules for unknown degradation.

In this article, we propose a two-step blind super-resolution method, namely, TSBSR, with transfer learning and an unsupervised scheme. First, we introduce transfer learning to address the problem of insufficient data of HSI, especially for remotely sensed HSI, by transferring the spatial knowledge from the remotely sensed color image domain to the HSI domain. This is achieved with a pretrained super-resolution model on the remotely sensed color images and a predefined degradation. Then, the super-resolution model is transferred to super-resolve the LR HSI band-by-band. The resulting pseudo-HR HSI restores the spatial information to some extent, but it neglects the spectral correlations and the difference between the predefined degradation and the real one. Therefore, we further present an unsupervised network with a learnable degradation prior to refining the pseudo-HR HSI. In particular, our unsupervised scheme decomposes LR and pseudo-HR HSIs into two sets of spectral meaningful basis and corresponding coefficients (i.e., the endmembers and abundances) to regularize the spectral characteristic between the LR and HR HSIs. Besides, our unsupervised scheme is augmented with a degradation prior via a learnable convolution layer to alleviate the spatial domain gap caused by predefined degradation. Extensive experiments demonstrate that our method can achieve better performance and relieve the problem of data-insufficient on remotely sensed HSI datasets, compared with the state-of-the-art single image super-resolution methods.

Our main contributions are threefold as follows.

1) We propose a two-step method with transfer learning and an unsupervised scheme for blind single remotely sensed HSI super-resolution, which can generalize to various unknown degradations without HSI training data.
2) We introduce transfer learning to address the problem of insufficient remotely sensed HSI training data by transferring spatial knowledge from the remotely sensed color image domain to the HSI domain.
3) We present an unsupervised scheme that is augmented with a learnable degradation prior to alleviating the spatial and spectral domain gaps of the pseudo-HR HSI produced by the transfer learning step.

The remainder of this article is organized as follows. In Section II, the related techniques of image super-resolution are listed. Section III describes the details of our method for blind HSI super-resolution. The experiment results and analysis are presented in Sections IV and V provides concluding remarks.

II. Related Work

In this section, we first review the nonblind single HSI super-resolution methods that are based on supervised DNN. Second, DNN-based unsupervised methods for blind HSI super-resolution are described. Third, we review some single HSI super-resolution methods based on spectral unmixing. Finally, some blind super-resolution technologies for color images and remotely sensed color images are introduced.

A. Supervised Nonblind Single HSI Super-Resolution

With the rapid development of deep learning, many DNN-based methods have achieved great success in a single HSI super-resolution field. Hu et al. [29] introduced a spectral difference DNN to fully explore the spectral and spatial correlations, where the spectral difference mapping between LR and HR HSIs was learned. Apart from the efforts made in network design, Xie et al. [30] embedded the nonnegative matrix factorization paradigm into a DNN, which well estimated the endmember and abundance counterparts. Recently, a grouped DNN with a progressive upsampling framework was proposed by Jiang et al. [14], named spatial-spectral prior for super-resolution (SSPSR). Such an architecture can extract spatial information while exploring the correlation cross spectra. Li et al. [16] pointed out that exploring the relationship between 2-D and 3-D convolutions was of great importance and developed a network named ERCSR to alternately employ 2-D and 3-D units. The entry of ERCSR is conventional 2-D convolutions, and the number of bands is fixed for a single ERCSR model. The most common critical limitation of these methods is their inability to handle an arbitrary number of bands.

Recently, some 3-D convolutional networks were developed. Mei et al. [31] replaced 2-D convolutions with 3-D ones, which can well extract both spatial and spectral information. Wang et al. [17] proposed a dual-channel network through 2-D and 3-D convolution to jointly exploit the information from both single band and adjacent bands, which can handle various HSI whose number of channels varies. Fu et al. [15] designed a bidirectional 3-D quasi-recurrent neural network (Bi-3DQRNN) in an encoder–decoder framework that explored structural spatial-spectral correlation and global correlation along spectra.

These DNN-based supervised methods require labeled training samples and precise degradation from the HR to LR HSIs. Once the degradation changes, all these methods need to be retrained to fit the new degradation. Moreover, the real degradation during the capture process is difficult to obtain, limiting the practical applications.

B. Unsupervised Blind HSI Super-Resolution

Recently, the problem of unknowing the precise degradation in hyperspectral imaging sensors has attracted more attention. Existing methods usually require an additional HR reference image (e.g., color image or multispectral image) as it provides spatial meaningful information. Qu et al. [32] proposed an unsupervised sparse Dirichlet-Net that naturally incorporated the two physical constraints of HSI and multispectral image. It was composed of two encoder–decoder networks and encouraged the representations from both modalities to follow [33]. Wang et al. [23] presented a deep-learning-based method, where the estimation of the observation model and fusion process was optimized iteratively and alternatingly during the super-resolution reconstruction. Liu et al. [25] introduced an implicit autoencoder network that integrated the nonnegative matrix factorization of the target HR HSI.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

These blind super-resolution methods require an auxiliary HR reference image that is precisely aligned with the input LR HSI since the HR reference image provides spatial meaningful information and LR HSI offers correct spectral details.

However, the requirements of multiple observations for the same scene with the observed LR HSI are unrealistic, thus limiting its application.

C. HSI Super-Resolution Based on Spectral Unmixing

To remove the constraint of auxiliary HR images that are aligned with LR observations, some researches [26], [27], [28] applied a pretrained single HSI super-resolution model to generate a coarse HR HSI, then optimized the coarse HR HSI in traditional manners.

Yuan et al. [26] trained a super-resolution model (SRDNN [34]) on an RGB dataset and then applied it to HSIs. Furthermore, there was a traditional nonnegative matrix factorization for alleviating spectral distortion. He and Liu [27] followed the idea in [26] and replaced SRDNN with a novel deep Laplacian pyramid network and used dictionary learning. Zou and Huang [28] designed a residual network on an RGB dataset, then a total variation was applied to encourage spectral information saving. Lu et al. [35] proposed a spatial correlation-regularized unmixing neural network for predicting the LR and HR abundance matrices. They first generate a coarse HR HSI by Bicubic interpolation [36], then generate the initial endmember by vertex component analysis [37].

However, these methods are designed to address nonblind super-resolution problems and lack specialized modules for unknown degradations. In addition, they usually recover the distorted spectral information by handcraft priors, which introduces several empirical parameters to tune. In this work, we present a blind super-resolution method for a single HSI, which can generalize to various unknown degradations without HSI training data. Besides, the traditional nonnegative matrix factorization is completed by a simple DNN in the proposed unsupervised scheme.

D. Blind Color Image Super-Resolution

Blind super-resolution of color images has been studied for more than ten years. Some methods [38], [39] decompose blind super-resolution into two steps, i.e., estimating the blur kernel from the given LR image, and then reconstructing the super-resolved image based on the estimated blur kernel. Michaeli and Irani [38] proposed internal patch recurrence to estimate the blur kernel. Gu et al. [39] trained a kernel estimation network and a correction network to iteratively correct the kernel. After that, Luo et al. [40] pointed out that the above two-step solution involves two independently trained models, thus adopting an alternating optimization algorithm that can estimate the blur kernel and reconstruct the HR image in a single model. In the remote sensing filed, Kang et al. [41] presented a multilayer degradation representation-guided blind super-resolution method. Liu et al. [42] proposed a graph neural network to consider the self-similarity patches in remotely sensed color imagery with a dual learning strategy.

However, these methods are mainly designed for blind super-resolution of color images and ignore the correlation cross spectra. Their performances are usually poor in maintaining the spectral curve, compared to those methods that are specifically designed for HSIs.

III. TSBSR Framework

In this section, we describe the proposed TSBSR framework for single remotely sensed HSI, which can be generalized to various degradations without HSI training data. Our TSBSR framework consists of a coarse super-resolution step via transfer learning and a refinement step via an unsupervised scheme.

A. Problem Formulation and Motivation

Given the HR and LR HSIs, denoted as $\mathbf{X} \in \mathbb{R}^{B\times M}$ and $\mathbf{Y} \in \mathbb{R}^{B\times m}$, respectively, where B, M, and m (with $m \ll M$) represent the number of spectral bands, number of pixels in the spatial domain for HR and LR HSIs, respectively. According to the degradation model, the relationship between LR and HR HSIs can be mathematically formulated as

$$\begin{aligned}\mathbf{Y} &= \mathbf{XA} + \mathbf{N} \\ &= (\mathbf{X} * \mathbf{k}) \downarrow_s + \mathbf{N}\end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{M\times m}$ denotes the degradation from HR HSI to LR one and $\mathbf{N} \in \mathbb{R}^{B\times m}$ represents the noise during imaging. Specifically, the degradation process generally consists of a blurring operation followed by a downsampling operation. $\mathbf{k}$ denotes the blur kernel and $\downarrow_s$ indicates the downsampling operation in spatial domain with scaling factor s. We use $*$ to indicate the blurring operation in the spatial dimensions of $\mathbf{X}$.

Conventional DNN-based methods [14], [15], [16] typically rely on a substantial amount of training data, which can be challenging to obtain for remotely sensed HSI. Particularly, in cases where only one HSI is available in the dataset, some approaches [14], [15] resort to splitting the image into training and testing regions. However, this limited amount of data is often insufficient for training a well-generalized DNN, resulting in overfitting issues.

Furthermore, supervised DNN-based methods [14], [15], [16], [17] often rely on known degradation parameters, such as the blur kernel $\mathbf{k}$ and noise $\mathbf{N}$, to generate paired LR and HR training data. As a result, these methods may struggle to generalize well to various degradation scenarios [21], [22]. In order to address this blind super-resolution problem, where the blur kernel $\mathbf{k}$ and noise $\mathbf{N}$ are unknown, several blind HSI super-resolution methods [23], [24], [25], [32] have been proposed. Though many existing blind HSI super-resolution methods are unsupervised, they often require an additional HR color image or multispectral image that is accurately aligned with the input LR HSI. In summary, supervised DNN-based methods struggle to generalize to various degradations, while unsupervised blind super-resolution approaches usually rely on auxiliary HR reference images, resulting in limited effectiveness in solving the blind single HSI super-resolution problem.

As a blind image super-resolution method, the proposed TSBSR framework neither requires paired LR and HR HSIs as most single HSI super-resolution methods need nor does it

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

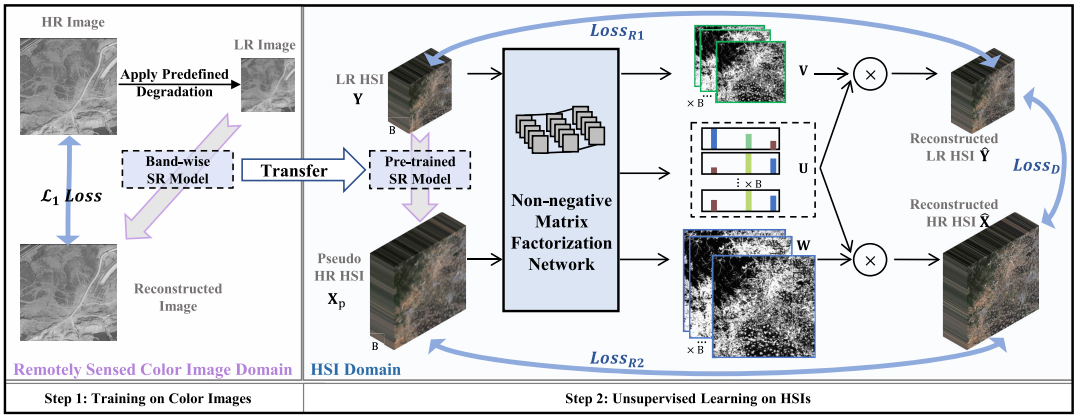


Fig. 1. Overview framework of the TSBSR method. First, we train a band-wise super-resolution model on numerous remotely sensed color images on Y channel (i.e., luminance), and the blur kernel is the Gaussian kernel. Second, we apply the pretrained super-resolution model on the LR input HSI, to generate a pseudo-HR HSI. To alleviate the gap between pseudo and latent HR HSIs, the LR input and pseudo-HR HSIs are fed into a nonnegative matrix factorization network. It iteratively decomposes both HSIs into spatial tensors ($\mathbf{V}$ and $\mathbf{W}$) and spectral tensor ($\mathbf{U}$). The pseudo-HR HSI and LR HSI share the spectral tensor $\mathbf{U}$ to force the same spectral information on both HSIs. Besides, a degradation prior is introduced to estimate the unknown degradation simultaneously. The whole unsupervised scheme minimizes Loss_{R1}, Loss_{R2}, and Loss_D. The final refined HSI is multiplied by $\mathbf{W}$ and $\mathbf{U}$ after enough iterations.

require an additional HR color image or multispectral image as most HSI blind super-resolution methods need. This makes our method more practical as it avoids the acquisition of paired LR–HR images or precisely aligned HR reference images, which is typically very challenging in practice. By utilizing our two-step design, our method has the potential to generalize to unknown degradation that may differ from the predefined degradations for which the model was trained.

The proposed TSBSR method for remotely sensed HSI super-resolution is shown in Fig. 1. In the first step, a band-wise super-resolution model is trained on a large dataset of remotely sensed color images, specifically on the Y channel (i.e., luminance) of the transformed YCbCr color space, as remotely sensed color images are more readily available and abundant compared to HSIs. This pretrained super-resolution model is then applied to the input LR HSI to generate a pseudo-HR HSI in a band-by-band manner.

However, there are two domain gaps between the pseudo and target HR HSIs, namely, *spectral gap* and *spatial gap*. Specifically, the resulting pseudo-HR HSI may suffer from distorted spectral information, as the band-wise super-resolution model treats each band of the HSI separately, neglecting the correlation along the spectral dimension. In addition, the pseudo-HR HSI is generated using a predefined Gaussian degradation, which may not generalize well to different degradation types [26]. To narrow both gaps, the second step of our TSBSR framework employs an unsupervised scheme to mitigate the issues, which utilizes a nonnegative matrix factorization network and a degradation prior.

B. Transfer Learning

Remotely sensed color images are more easily accessible compared to remotely sensed HSIs, and there exist numerous publicly available remotely sensed color image datasets that offer a substantial quantity and quality of images. In contrast, remotely sensed HSI datasets are relatively scarce. In the remote sensing community, some public datasets contain only one remotely sensed HSI, and some studies [14], [15] employ training and testing on distinct regions within the same image. However, this practice can potentially lead to severe overfitting due to the inherent similarity between the testing and training regions, which cannot be avoided. Furthermore, previous researches [26], [27], [28] have shown that transferring knowledge from color images to the HSI domain can greatly improve representation and generalization ability. Therefore, we incorporate transfer learning into our method by transferring spatial knowledge from the remotely sensed color image domain to the HSI domain.

During the transfer learning step, HSI training data is not required. We train the super-resolution model using a large dataset of remotely sensed color images from the AID dataset [43], instead of remotely sensed HSIs. The AID dataset [43] contains 10 000 remotely sensed color images with a resolution of $600 \times 600 \times 3$. The super-resolution model, initially trained on three-channel remotely sensed color images, cannot be directly applied to remotely sensed HSIs with different numbers of bands. Thus, we convert the remotely sensed color images to the YCbCr color space and extract the Y channel (luminance channel) [34] to train a single-channel super-resolution model.

For the training of our super-resolution model on the AID dataset [43], we choose the robust single image super-resolution model called SwinIR [44]. Since the pretrained model is based on supervised learning, paired LR–HR training samples are required. The degradation is defined in advance [15], [24], [45]. Specifically, we first apply an 8×8 Gaussian filter with a standard deviation of 3 to smooth the Y channel of the remotely sensed color image. Then, we downsample the Y channel using the Bicubic resampling [36] with a specific scale factor. Finally, each Y channel is corrupted by zero-mean Gaussian noise with a random selection of σ from 2.5 to 10, corresponding to adding Gaussian noise into the Y channel until the signal-to-noise ratio (SNR) reaches 30–40 dB. Following the training of the SwinIR model on the AID dataset using the predefined degradation, it is employed to process the input LR HSI. The model generates a pseudo-HR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

HSI by processing each band individually, and then the resulting bands are concatenated along the spectral dimension.

C. Unsupervised Scheme

The first step of our framework aims at obtaining a pseudo-HR HSI by a band-wise super-resolution model that is pretrained on remotely sensed color images instead of HSIs with a predefined degradation. This provides decent initial results and overcomes the difficulties of acquiring a large amount of remotely sensed HSIs for training. Despite that, it does cause the train-test distribution gap, which we seek to address in the second step. Specifically, these two gaps can be summarized as the spectral domain gap and the spatial domain gap. The spectral gap arises from the band-wise super-resolution model that neglects the correlation along spectral. The spatial domain gap is caused by the predefined degradation, while the real degradation is unable to obtain in advance.

The second step of our method is the key to alleviating the train-test distribution gaps. To achieve it, we propose an unsupervised scheme that takes the input LR and pseudo-HR HSI as input and returns the refined HR HSI. This scheme is made of two essential modules, i.e., a nonnegative matrix factorization network for alleviating the spectral domain gap and a degradation prior to reducing the spatial domain gap.

Next, we provide a detailed description of the working mechanisms of both modules.

1) Nonnegative Matrix Factorization Network: The nonnegative matrix factorization network reduces the spectral domain gap by factorizing the HSI into two components where one component is correlated with spectral information (known as endmember matrix) and the other one is correlated with spatial information (known as abundance matrix). Specifically, we first factorize both the input LR HSI and pseudo-HR HSI into endmember and abundance matrices. Then, we build the refined HR HSI with the spatial information (abundance) of the pseudo-HR HSI and the spectral information (endmember) of the input LR HSI. Since the refined HR HSI is built with the spectral information of input LR HSI, the spectral domain gap could be reduced.

Formally, an HSI could be factorized into a combination of endmember and abundance matrices, based on the theory of nonnegative matrix factorization [46]. As a result, the LR HSI $\mathbf{Y}$ could be formulated as

$$\mathbf{Y} = \mathbf{UV} \tag{2}$$

where $\mathbf{U} \in \mathbb{R}^{B\times L}$ is the endmember matrix representing the number of spectral signatures of the underlying materials and L is the number of endmembers. $\mathbf{V} \in \mathbb{R}^{L\times m}$ denotes the abundance matrix whose column vector represents the fractions of all endmembers within each pixel of the scene.

Since the pseudo-HR HSI $\mathbf{X}_p$ is super-resolved from the LR HSI $\mathbf{Y}$, their scenes are the same. Therefore, their underlying materials should be the same as well, which means that they should share the same endmember matrix $\mathbf{U}$, as shown in Fig. 1. Accordingly, the pseudo-HR HSI $\mathbf{X}_p$ could be approximated as

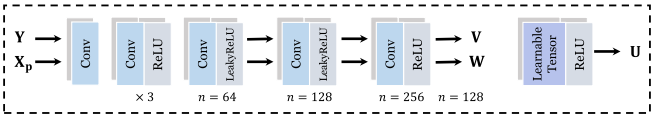


Fig. 2. Architecture of nonnegative matrix factorization network.

$$\mathbf{X}_p = \mathbf{UW} \tag{3}$$

where $\mathbf{U}$ is the endmember of the observed LR HSI and $\mathbf{W} \in \mathbb{R}^{L\times M}$ denotes the abundance matrix of pseudo-HR HSI.

As a result, the objective function of the nonnegative matrix factorization could be formulated as

$$\begin{aligned} &\arg\min \ \|\mathbf{Y} - \mathbf{UV}\|_F^2 \ + \|\mathbf{X}_p - \mathbf{UW}\|_F^2 \\ &\text{s.t. } \mathbf{U} \geq 0, \quad \mathbf{V} \geq 0, \ \mathbf{W} \geq 0 \\ &\text{s.t. } \mathbf{V}^T 1_L = 1_m, \quad \mathbf{W}^T 1_L = 1_M \end{aligned} \tag{4}$$

where we apply an extra nonnegative constraint on the endmember and abundance tensors, and a sum-to-one constraint on abundance [26].

Instead of adopting traditional optimization methods for the nonnegative matrix factorization, we propose a nonnegative matrix factorization network to optimize the objective function, as shown in Fig. 2. The inputs are LR HSI $\mathbf{Y}$ and pseudo-HR HSI $\mathbf{X}_p$, and the outputs are abundance matrices $\mathbf{V}$ and $\mathbf{W}$. The proposed nonnegative matrix factorization network consists of several convolutional layers, and the final rectified linear unit (ReLU) layer is utilized to satisfy the nonnegativity of abundance matrices. The number of features may change from different layers, and we use n to represent the number of features. The endmember $\mathbf{U}$ in our nonnegative matrix factorization network is defined as a learnable tensor and its shape is $B \times L$, where B denotes the number of bands and L is the number of endmembers. To make sure that the endmember $\mathbf{U}$ is nonnegative, we use a ReLU function following the learnable tensor $\mathbf{U}$.

In the proposed nonnegative matrix factorization network, we introduce two loss functions, namely, Loss_{R1} and Loss_{R2}, to constraint the outputs of the network, so that it minimizes the optimization objective in (4).

The loss function Loss_{R1} is the L_1 loss between the reconstructed LR HSI $\hat{\mathbf{Y}} = \mathbf{UV}$ and input LR HSI $\mathbf{Y}$, which can be expressed as

$$\text{Loss}_{R1}\left(\hat{\mathbf{Y}}, \mathbf{Y}\right) = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{Y} - \mathbf{UV}| \tag{5}$$

where N denotes the training batch size. $\mathbf{U}$ and $\mathbf{V} = f_\Theta(\mathbf{Y})$ are the endmember and abundance estimated by the nonnegative matrix factorization network $f_\Theta(\cdot)$.

The Loss_{R2} is designed to narrow the differences between the reconstructed HR HSI $\hat{\mathbf{X}} = \mathbf{UW}$ and pseudo-HR HSI $\mathbf{X}_p$, which is formulated as

$$\text{Loss}_{R2}\left(\hat{\mathbf{X}}, \mathbf{X}_p\right) = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{X}_p - \mathbf{UW}| \tag{6}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{W} = f_\Theta(\mathbf{X}_p)$ is the abundance estimated by the nonnegative matrix factorization network $f_\Theta(\cdot)$.

In order to simulate the iterative optimization process in (4), all parameters of estimating abundance matrices $\mathbf{V}$ and $\mathbf{W}$ are shared.

2) Degradation Prior: The degradation prior narrows the spatial domain gap that is caused by the predefined degradation while training the SwinIR model. In Fig. 1, we can see that there is an implicit relationship between the reconstructed LR HSI $\hat{\mathbf{Y}}$ and the reconstructed HR HSI $\hat{\mathbf{X}}$. The refined super-resolution result $\hat{\mathbf{X}}$ can be degraded to the LR version, and it ought to be similar to the LR HSI $\hat{\mathbf{Y}}$. We adopt a learnable convolution layer with kernel $\mathbf{k}$ and a Bicubic downsampling to model the structure of degradation $\mathbf{A}$, which equals inference of the unknown blur kernel $\mathbf{k}$. Furthermore, the learnable convolution layer is initialized by a zero-mean Gaussian distribution, which proves to be an effective prior for blur kernel [24].

Hence, we introduce a degradation loss, namely, Loss_D, to regularize the difference between the degraded $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$. Formally, the Loss_D can be expressed as

$$\text{Loss}_D(\hat{\mathbf{X}}, \hat{\mathbf{Y}}) = \frac{1}{N}\sum_{i=1}^{N}\left|\mathbf{Y} - \left(\hat{\mathbf{X}} * \mathbf{k} \downarrow_s\right)\right| \tag{7}$$

where $\hat{\mathbf{X}} * \mathbf{k} \downarrow_s$ denotes the degraded version of $\hat{\mathbf{X}}$ is blurred by a convolution with kernel $\mathbf{k}$ and then downsampled by a factor of s.

As a consequence, the total losses of our unsupervised network can be expressed as

$$\text{Loss} = \text{Loss}_{R1} + \text{Loss}_{R2} + \eta\text{Loss}_D \tag{8}$$

where η is the balance coefficient.

During the optimization of the learnable convolution layer, we jointly optimize the reconstructed HR HSI with the learned degradation to meet the constraint between the degraded $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ as well, so that the spatial domain gap could be reduced.

IV. Experiments

In this section, we conduct a series of experiments on seven remotely sensed HSI datasets. First, we introduce the experiment datasets and settings. Second, the quantitative and qualitative results of competing methods and ours are listed. Third, we conduct experiments on real remotely sensed HSI data. Finally, we provide the ablation studies to demonstrate the effectiveness of the proposed modules in our approach.

A. Datasets and Settings

1) Datasets: We evaluate seven remotely sensed HSI datasets, including Pavia Centre and University [47], Washington DC Mall (WDC), Salinas, Kennedy Space Center (KSC) [9], Botswana, Houston-2013 [48], and Houston-2018 [49]. Our TSBSR method is based on unsupervised learning and requires no HSI training data. The band-wise SwinIR model [44] in the transfer learning step is trained on a remotely sensed color image dataset, i.e., AID dataset [43].

a) Pavia Centre and University are two scenes from Pavia, Northern Italy, acquired by the ROSIS sensor during a flight campaign. The spatial resolution of Pavia Centre is 1096 × 715, and the number of spectral bands is 102. The spatial resolution of Pavia University is 610 × 340, and it has 103 bands, which is one more band than Pavia Centre. **b) WDC** is provided by the Spectral Information Technology Application Center of Virginia. The spatial resolution is 1208 × 307, with 191 spectral bands. **c) Salinas** is collected by the AVIRIS sensor over the Salinas Valley, CA, USA. The spatial resolution is 512 × 217, with 224 spectral bands. **d) KSC** is collected by the AVIRIS sensor over the KSC, FL, USA. The spatial resolution is 512 × 614, with 176 spectral bands. **e) Botswana** is collected by the NASA EO-1 satellite over the Okavango Delta, Botswana. The spatial resolution is 1476 × 256, with 145 spectral bands. **f) Houston-2013** is acquired over the University of Houston campus and the neighboring urban area. The spatial resolution is 349 × 1905, and has 144 spectral bands in the 380–1050 nm region. **g) Houston-2018** covers a 380–1050 nm spectral range with 48 bands, and the spatial resolution is 1202 × 4172.

2) Competing Methods and Metrics: We compare with seven typical state-of-the-art super-resolution methods, including A+ [50], NLRTATV [13], TLSR [26], SSPSR [14], ERCSR [16], Bi-3DQRNN [15], and DLGNN [42].

The competing methods can be divided into two types, i.e., methods that require HSI training data and methods that require no HSI training data. The former type is mainly traditional methods that can be applied to testing HSIs directly, such as A+, NLRTATV, and TLSR. The latter type is learning-based methods that need to be trained on the training dataset, including SSPSR, ERCSR, Bi-3DQRNN, and DLGNN.

A+ is a powerful traditional color image super-resolution method and can be utilized to enhance HSIs by considering each HSI as a sequence of grayscale images. NLRTATV is a traditional HSI super-resolution method that can be applied to testing data directly. TLSR requires both a pseudo-HR HSI and an LR HSI as inputs. The pseudo-HR HSI is generated using our pretrained SwinIR.

SSPSR and ERCSR are 2-D DNN-based methods that each model can only handle HSIs with a fixed number of bands. As for Bi-3DQRNN, it is a 3-D DNN-based method, which can tackle a various number of bands during training and testing. DLGNN is a blind super-resolution method for remotely sensed color images. These four methods are supervised and trained on paired LR–HR HSIs and the LR HSI is generated by a Gaussian blur kernel and downsampling. During training, noise with random strengths is added. Specifically, SSPSR, ERCSR, Bi-3DQRNN, and DLGNN are trained on the predefined degradation, i.e., a 8 × 8 Gaussian blur kernel and a Bicubic downsampling [36], and each HSI is corrupted by zero-mean Gaussian noise with random intensities σ is randomly selected from 2.5 to 10. During testing, the degradations, including blur kernel and noise level, are various.

Evaluation indicators are PSNR, SSIM [51], and SAM [52], to evaluate image quality, image similarity, and vector similarity, respectively. PSNR and SSIM are two popular metrics to evaluate spatial fidelity, while SAM is based on spectra and can be used to measure the spectral similarity of the ground

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

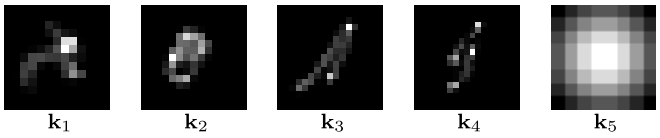


Fig. 3. Five blur kernels utilized for degradation in the spatial domain.

truth $\mathbf{X}$ and the reconstructed image $\hat{\mathbf{X}}$. The larger values of PSNR and SSIM, and the smaller value of SAM, suggest better performance.

3) Experiment Settings: In our experiment, we first train the SwinIR model [44] for the predefined degradation on plenty of remotely sensed color images, to generate the pseudo-HR HSI. Then, we use the nonnegative matrix factorization network to estimate the endmember and abundance tensors of the LR and pseudo-HR HSIs. After a certain number of iterations, the endmember and abundance of the pseudo-HR HSI are composed to reconstruct the final HR HSI by matrix multiplication. Our method is conducted with the deep-learning framework PyTorch with NVIDIA GeForce RTX 3090 GPU.

a) Pretrained super-resolution model: Our pretrained super-resolution model on the remotely sensed color image domain is learned by minimizing the $\mathcal{L}_1$ loss between the predicted HR image and the ground truth during the training phase. The predefined degradation consists of a Gaussian blur kernel, Bicubic downsampling, and zero-mean Gaussian noise. Specifically, we first use an 8×8 Gaussian filter (standard deviation is 3) to smooth the Y channel of remotely sensed color images. Then, we downsample each Y channel by a scale factor with Bicubic resampling. Each image is corrupted by zero-mean Gaussian noise with random intensities σ is randomly selected from 2.5 to 10. The learning rate is initialized as 1×10^{-4}, and the minimum learning rate is set to 1×10^{-5}. The network parameters are optimized using the AdamW optimizer [53].

b) Unsupervised scheme: The parameters in (8) are set $\eta = 2$ for the first 50 epochs and $\eta = 1$ for the following epochs. Similar to the pretrained super-resolution model, we also adopt $\mathcal{L}_1$ loss and the learning rate is initialized as 1×10^{-3}, and the minimum learning rate is set to 5×10^{-5}. Five different blur kernels are shown in Fig. 3, and the size of them are 13×13, 15×15, 17×17, 19×19, and 8×8, respectively. Besides, there are three different levels of noise corruption, i.e., $\sigma = 2.5, 5, 10$.

To demonstrate the performances of competing methods and ours under unknown degradation, we set three variables during experiments, they are five blur kernels $(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4, \mathbf{k}_5)$, three noise levels $(\sigma = 2.5, 5, 10)$, and two-scale factors $(\times 4, \times 8)$. Once one of the variables is changed, the other two variables are fixed. Since we apply the Gaussian kernel and randomly noises to simulate the degradation, the test blur kernels $(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4, \mathbf{k}_5)$ and noise levels $(\sigma = 2.5, 5, 10)$ are unknown during training.

B. Experiments on Remotely Sensed HSI Data

In this section, we conduct experiments on seven remotely sensed HSI datasets. Our TSBSR method requires no HSI training data, the pretrained super-resolution model SwinIR is trained on a lot of remotely sensed color images, i.e., the AID dataset. The way of generating LR–HR training pairs is the same as other supervised learning methods. Considering the specificity of remotely sensed HSI datasets, i.e., only one image in each dataset, we choose and crop the testing regions from each remotely sensed dataset. In the case of supervised HSI super-resolution competing methods, the remaining regions are utilized for training. As for Pavia Centre and University dataset, we choose Pavia Centre as the training set and test on Pavia University. Since the number of bands in Pavia Centre is 102 and Pavia University is 103, we drop the last band in Pavia University while testing. The training set is built by following [15], and each training sample has a spatial size of 64×64 and a full spectral size. Besides, some data augmentation approaches such as rotating are employed to generate more training samples. As for testing, we crop the center region at the resolution of 340×340 in Pavia University and Houston-2013, 304×304 in WDC, 128×128 in Salinas, 240×240 in KSC and Botswana, and 512×512 in Houston-2018. More details can be found in Table I.

TABLE I
DETAILS OF TESTING REGION SIZE AND POSITION IN EACH REMOTELY SENSED HSI. IN THE *Position* COLUMN, WE USE THE TWO COORDINATES TO REPRESENT THE TOP LEFT AND BOTTOM RIGHT VERTEXES, RESPECTIVELY

Dataset	Size	Position
Pavia University	340×340	(0, 135) & (339, 474)
WDC	304×304	(452, 1) & (755, 304)
Salinas	128×128	(192, 45) & (319, 172)
KSC	240×240	(136, 187) & (375, 426)
Botswana	240×240	(618, 8) & (857, 247)
Houston-2013	340×340	(4, 782) & (343, 1121)
Houston-2018	512×512	(345, 1830) & (856, 2341)

Performance on Pavia University Dataset: Here, we evaluate the performance of the proposed method on the Pavia University dataset under different degradation with various blur kernels, noise levels, and scale factors.

1) Various Blur Kernels: We utilize five kernels shown in Fig. 3 with a fixed noise level (i.e., $\sigma = 10$) of noise corruption and a fixed scale factor $\times 4$ to produce different degradations. The size of blur kernels $\mathbf{k}_{1-5}$ are 13×13, 15×15, 17×17, 19×19, and 8×8, respectively.

The super-resolution results of reconstructing Pavia University are illustrated in Table II. We can see that our method obtains excellent results among all blur kernels. NLRTATV performances the worst for all kernels, as it is a traditional method and relies on handcraft priors. A+ can obtain higher values of PSNR and SSIM than NLRTATV, but worse SAM values. It is reasonable because A+ is a powerful color image super-resolution method while NLRTATV investigates spectral information. Focusing on color image super-resolution methods, i.e., A+ and DLGNN, neglect the correlation across spectra, according to the SAM metric. Referring to PSNR and SSIM, DLGNN can obtain higher spatial fidelity compared with A+, because DLGNN is based on DNNs and A+ is a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

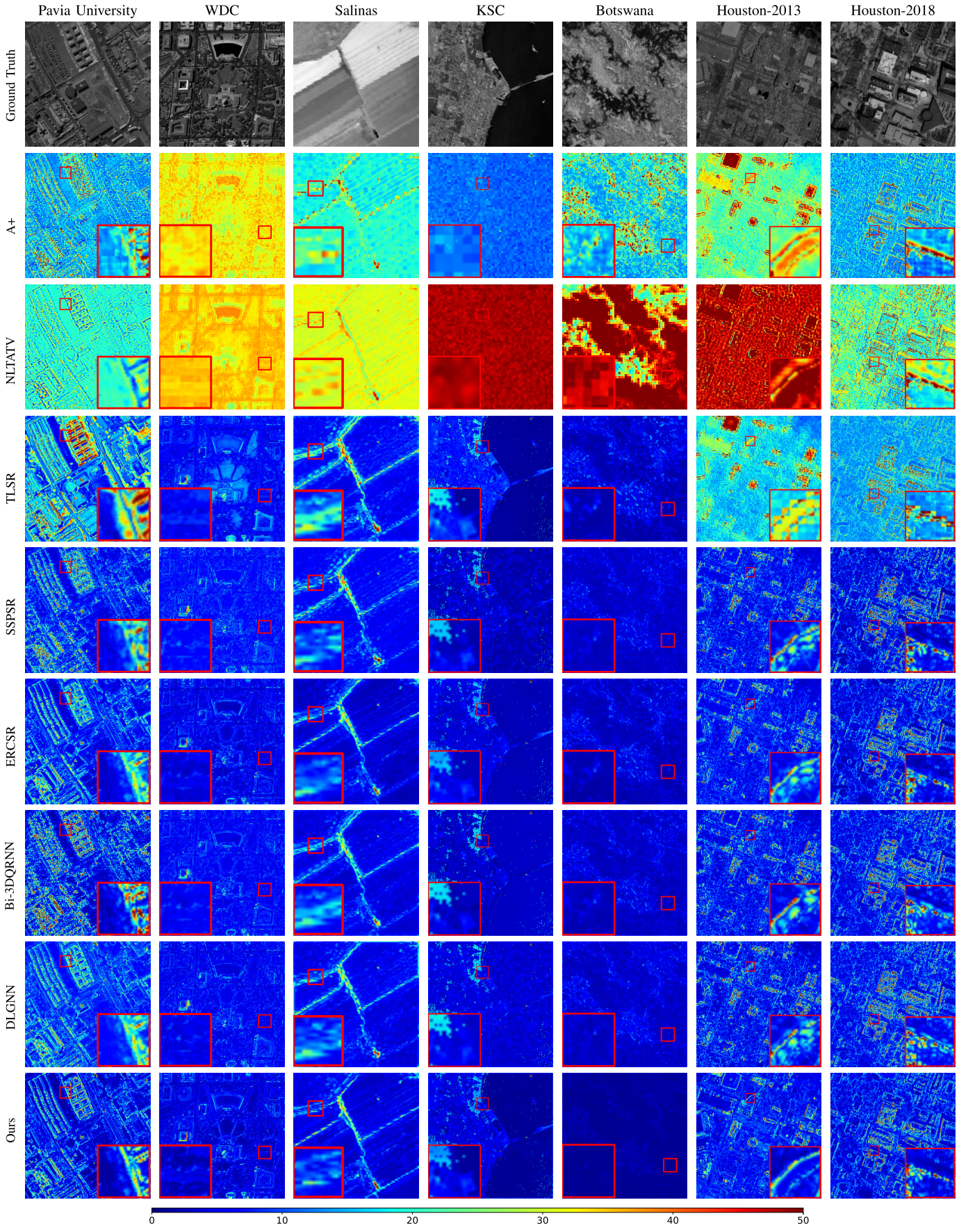


Fig. 4. HSI blind super-resolution results on (from the left to the right) Pavia University, WDC, Salinas, KSC, Botswana, Houston-2013, and Houston-2018 datasets under the degradation with blur kernel $\mathbf{k_1}$ and $\sigma = 10$, and the scale factor is 4.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

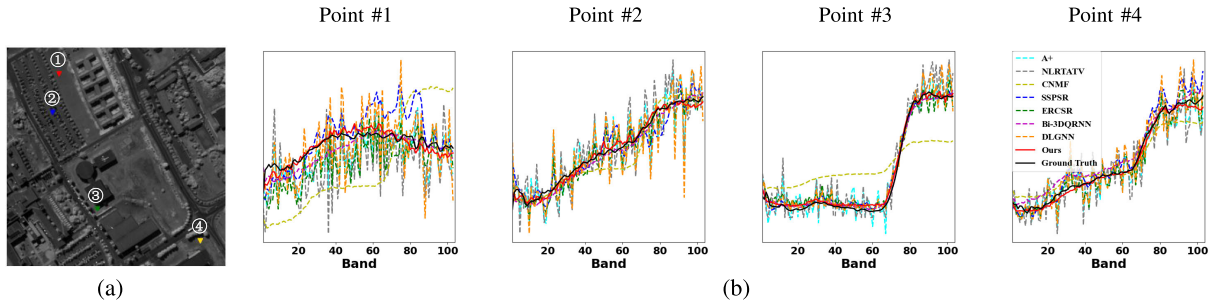


Fig. 5. Spectral visualization on real data. (a) Test image, i.e., Pavia University. (b) Recovered spectra on the four patches indicated in (a).

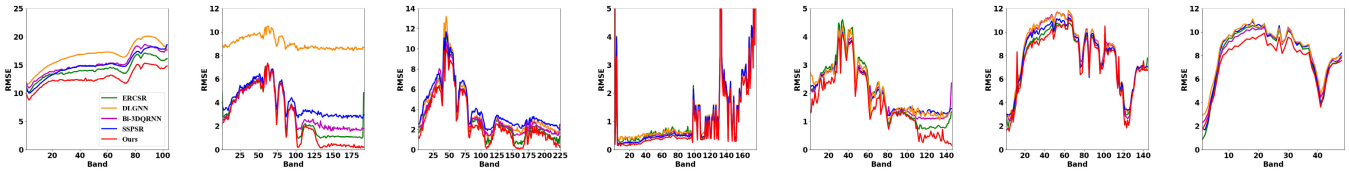


Fig. 6. RMSE curves on Pavia University, WDC, Salinas, KSC, Botswana, Houston-2013, and Houston-2018 datasets under the degradation with blur kernel $\mathbf{k_1}$ and $\sigma = 10$, and the scale factor is 4.

TABLE II

QUANTITATIVE EVALUATION OF ALL COMPETING METHODS WHEN THE NOISE LEVEL σ IS 10, THE SCALE FACTOR IS $\times 4$, AND THE BLUR KERNEL IS UNKNOWN (I.E., $\mathbf{k}_1 - \mathbf{k}_5$) BY AVERAGE PSNR/SSIM/SAM ON PAVIA UNIVERSITY DATASET

Kernels	Metrics	Methods							
		A+ [50]	NLRTATV [13]	TLSR [26]	SSPSR [14]	ERCSR [16]	Bi-3DQRNN [15]	DLGNN [42]	TSBSR (Ours)
$\mathbf{k}_1$	PSNR ↑	22.3492	21.3063	23.0794	24.7764	25.1384	23.7055	24.5559	**26.1218**
	SSIM ↑	0.5553	0.5919	0.7396	0.7443	0.7429	0.7333	0.7144	**0.7944**
	SAM ↓	0.4176	0.3795	0.3548	0.1853	0.1800	0.1492	0.2340	**0.1221**
$\mathbf{k}_2$	PSNR ↑	22.1560	21.2044	23.3423	26.1656	25.3966	25.7591	24.8763	**26.6918**
	SSIM ↑	0.5508	0.5472	0.7511	0.7910	0.7525	0.8013	0.7269	**0.8084**
	SAM ↓	0.4255	0.4419	0.3445	0.1631	0.1806	0.1308	0.2300	**0.1161**
$\mathbf{k}_3$	PSNR ↑	21.8359	20.3964	22.7255	24.1946	24.1930	23.9304	23.6866	**24.9631**
	SSIM ↑	0.5297	0.4972	0.7184	0.6981	0.7011	0.7048	0.6650	**0.7366**
	SAM ↓	0.4430	0.4499	0.3719	0.1823	0.1902	0.1497	0.2418	**0.1406**
$\mathbf{k}_4$	PSNR ↑	21.9614	20.2588	23.1638	25.3505	24.6869	25.4376	24.2732	**25.6005**
	SSIM ↑	0.5361	0.5222	0.7350	0.7473	0.7191	0.7568	0.6915	**0.7587**
	SAM ↓	0.4361	0.4288	0.3529	0.1660	0.1841	0.1311	0.2344	**0.1305**
$\mathbf{k}_5$	PSNR ↑	24.7362	22.7852	23.5385	26.1563	25.9802	26.2989	25.3196	**26.6040**
	SSIM ↑	0.6918	0.5826	0.7389	0.7708	0.7656	0.7912	0.7462	**0.7978**
	SAM ↓	0.3039	0.3745	0.2540	0.1517	0.1731	0.1303	0.1921	**0.1152**

traditional method. TLSR method uses the same pseudo-HR HSI as our TSBSR approach, but the performance of TLSR is limited due to the traditional way of refining the pseudo-HR HSI. As for the DNN-based HSI super-resolution methods, i.e., SSPSR, ERCSR, and Bi-3DQRNN, the performances of them are unstable among five blur kernels. We can see that ERCSR reaches the best PSNR in $\mathbf{k}_1$ kernel degradation, SSPSR performs best in $\mathbf{k}_2$ and $\mathbf{k}_3$ kernel degradations, and Bi-3DQRNN is the highest PSNR in $\mathbf{k}_4$ kernel degradation. Besides, Bi-3DQRNN reaches the best SAM metric among all competing methods. In contrast to the experiments conducted on blur kernels $\mathbf{k_{1-4}}$, the reconstructed results of the competing methods on the Gaussian kernel $\mathbf{k}_5$ are better. The reason is that the competing methods are all trained on the Gaussian blur kernel, which is the same as the testing kernel.

2) Various Noise Levels: We utilize three noise levels ($\sigma = 2.5, 5, 10$) with a fixed blur kernel $\mathbf{k}_1$ and scale factor $\times 4$ to produce different degradations.

The results of various noise corruptions are shown in Table III, we can observe that our method surpasses all competing methods with a clear margin in all criteria, especially for noise level $\sigma = 10$. The noise level is blind during training, we adopt randomly noises between $\sigma \in [2.5, 10]$. Since our TSBSR method takes the corrupted HSI as input

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

TABLE III

QUANTITATIVE EVALUATION OF ALL COMPETING METHODS WHEN THE BLUR KERNEL IS $\mathbf{k}_1$, SCALE FACTOR IS $\times 4$, AND THE NOISE LEVEL IS UNKNOWN (I.E., $\sigma = 2.5, 5, 10$) BY AVERAGE PSNR/SSIM/SAM ON PAVIA UNIVERSITY DATASET

Sigmas	Metrics	Methods							
		A+ [50]	NLRTATV [13]	TLSR [26]	SSPSR [14]	ERCSR [16]	Bi-3DQRNN [15]	DLGNN [42]	TSBSR (Ours)
2.5	PSNR ↑	24.6695	22.5215	22.7267	25.2199	25.0412	23.6680	26.0610	**26.2208**
	SSIM ↑	0.7711	0.7109	0.7301	0.7708	0.7906	0.7377	0.7794	**0.7987**
	SAM ↓	0.3165	0.2781	0.3668	0.1520	0.1312	0.1445	0.1285	**0.1209**
5	PSNR ↑	24.0880	22.1201	22.9831	25.1293	25.8378	23.6849	25.9037	**26.1419**
	SSIM ↑	0.7135	0.7084	0.7384	0.7659	0.7755	0.7372	0.7793	**0.7971**
	SAM ↓	0.3412	0.2856	0.3567	0.1598	0.1475	0.1456	0.1556	**0.1207**
10	PSNR ↑	22.3492	21.3063	23.0794	24.7764	25.1384	23.7055	24.5559	**26.1218**
	SSIM ↑	0.5553	0.5919	0.7396	0.7443	0.7429	0.7333	0.7144	**0.7944**
	SAM ↓	0.4176	0.3795	0.3548	0.1853	0.1800	0.1492	0.2340	**0.1221**

TABLE IV

QUANTITATIVE EVALUATION OF ALL COMPETING METHODS WHEN THE BLUR KERNEL IS $\mathbf{k}_1$ THE NOISE LEVEL $\sigma = 10$ BY AVERAGE PSNR/SSIM/SAM ON PAVIA UNIVERSITY DATASET FOR $\times 4$ AND $\times 8$ CASES

Scale Factors	Metrics	Methods							
		A+ [50]	NLRTATV [13]	TLSR [26]	SSPSR [14]	ERCSR [16]	Bi-3DQRNN [15]	DLGNN [42]	TSBSR (Ours)
$\times 4$	PSNR ↑	22.3492	21.3063	23.0794	24.7764	25.1384	23.7055	24.5559	**26.1218**
	SSIM ↑	0.5553	0.5919	0.7396	0.7443	0.7429	0.7333	0.7144	**0.7944**
	SAM ↓	0.4176	0.3795	0.3548	0.1853	0.1800	0.1492	0.2340	**0.1221**
$\times 8$	PSNR ↑	20.3829	19.9775	24.7230	23.6585	22.9731	23.6841	22.8441	**25.9110**
	SSIM ↑	0.4812	0.6737	0.7016	0.6923	0.6553	0.7035	0.6529	**0.7866**
	SAM ↓	0.5200	0.4020	0.3845	0.1767	0.2451	0.1531	0.2523	**0.1288**

for the guidance to refine the pseudo-HR HSI, our method is more adaptive to different noise levels, compared with the supervised DNN-based methods.

3) Different Scale Factors: We utilize two scale factors ($\times 4$ and $\times 8$) with a fixed blur kernel $\mathbf{k}_1$ and noise level $\sigma = 10$ to produce different degradations. Table IV shows the reconstruction results. We can see that our approach still surpasses all competitors with a clear margin, which is consistent with the former experiments.

In the first column of Fig. 4, the visual results of Pavia University under kernel $\mathbf{k}_1$, noise level $\sigma = 10$, and the scale factor $\times 4$ are shown. We use error maps to demonstrate the performance of each method, the deeper blue color means less error from the ground truth. We can clearly see that the proposed method recovers more sharp edges and fewer differences. In Fig. 5, we select four points from Pavia University and visualize their spectra of all methods. We represent the values of the ground truth with red lines, our method with black lines, and other methods with dashed lines. It can be observed that our result most closely matches the ground truth. For comparing the reconstructed result over all bands, the root mean square error (RMSE) is calculated between the ground truth and reconstructed HSI in Fig. 6. Because the performances of A+, NLRTATV, and TLSR are significantly worse, they are not plotted in the RMSE figure, for better visualizing the comparable methods. We can observe that the proposed method exhibits minimal error on the Pavia University dataset.

Performance on Other Remotely Sensed HSI Datasets: In this part, we focus on the test setting of blur kernel $\mathbf{k}_1$, noise level $\sigma = 10$, and scale factor $\times 4$. The datasets for evaluation are WDC, Salinas, KSC, Botswana, Houston-2013, and Houston-2018.

The quantitative results are shown in Table V. It can be seen that our method surpasses all competing methods among all datasets. One reason is that the number of training samples is real scarce, which is bad for data-driven methods. Compared with TLSR, the proposed TSBSR method can directly optimize the parameters by DNNs instead of hand-craft priors, thus reconstructing better in both super-resolution quality and spectral distortion. The state-of-the-art supervised methods perform well in super-resolution quality of HSIs with predefined degradation, but for blind super-resolution, they introduce too much spectral distortion, resulting in relatively poor performance on remotely sensed HSI. The reconstructed results of all datasets are shown in Fig. 4. The small target is zoomed in a red box. We observe that the NLRTATV method yields the worst results on all datasets. While A+ performs slightly better than NLRTATV, it still yields poor results. The remaining competing methods recover more details than traditional methods such as A+ and NLRTATV, but they are still limited at preserving fine-grained details. For example, TLSR struggles to recover the edges, while our method shows

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

TABLE V

QUANTITATIVE EVALUATION OF ALL COMPETING METHODS WHEN THE BLUR KERNEL IS $\mathbf{k}_1$ THE NOISE LEVEL $\sigma = 10$ BY AVERAGE PSNR/SSIM/SAM ON WDC, SALINAS, KSC, BOTSWANA, HOUSTON-2013, AND HOUSTON-2018 DATASETS FOR $\times 4$ SCALE

Datasets	Metrics	Methods							
		A+ [50]	NLRTATV [13]	TLSR [26]	SSPSR [14]	ERCSR [16]	Bi-3DQRNN [15]	DLGNN [42]	TSBSR (Ours)
WDC	PSNR ↑	24.1256	23.8267	34.9304	36.3944	41.3223	37.9255	38.6924	**44.5684**
	SSIM ↑	0.6101	0.6013	0.9433	0.9460	0.9657	0.9618	0.9624	**0.9657**
	SAM ↓	0.2679	0.6091	0.0256	0.0265	0.0197	0.0225	0.0213	**0.0188**
Salinas	PSNR ↑	28.2630	27.9435	33.4703	36.8850	36.4309	38.4774	38.1757	**41.5058**
	SSIM ↑	0.7859	0.7478	0.8705	0.9465	0.9339	0.9588	0.9565	**0.9643**
	SAM ↓	0.5419	0.4643	0.2067	0.0664	0.0777	0.0587	0.0600	**0.0525**
KSC	PSNR ↑	30.1221	29.8378	37.8802	50.4305	51.0388	50.7036	49.1251	**51.3710**
	SSIM ↑	0.6187	0.5712	0.9857	0.9914	0.9919	0.9926	0.9923	**0.9927**
	SAM ↓	0.9028	0.9989	0.7909	0.9290	0.6744	0.5179	0.8669	**0.3762**
Botswana	PSNR ↑	25.7939	23.1932	40.2908	43.4332	42.4604	42.7537	42.4602	**44.2439**
	SSIM ↑	0.5304	0.6797	0.9720	0.9724	0.9766	0.9748	0.9734	**0.9776**
	SAM ↓	0.9353	0.4069	0.2717	0.1344	0.1562	0.1460	0.1607	**0.1330**
Huston-2013	PSNR ↑	27.6709	20.1956	28.5532	30.9114	31.1774	30.4549	30.3133	**31.3179**
	SSIM ↑	0.7578	0.4738	0.7838	0.8662	0.8713	0.8644	0.8643	**0.8732**
	SAM ↓	0.3815	0.4398	0.2602	0.1575	0.1685	0.1536	0.1551	**0.1506**
Huston-2018	PSNR ↑	27.4670	25.8224	27.8185	30.3554	30.7968	30.2800	29.9978	**31.0437**
	SSIM ↑	0.7957	0.7261	0.8536	0.9005	0.9010	0.8928	0.8925	**0.9011**
	SAM ↓	0.2434	0.2913	0.1560	0.0838	0.0915	0.0881	0.0854	**0.0803**

clearly better reconstruction results on edges. In Fig. 6, we plot the error maps of the remaining datasets under kernel $\mathbf{k}_1$, noise level $\sigma = 10$, and the scale factor is $\times 4$. It can be seen that the proposed method exhibits minimal error in all datasets.

In brief, the traditional methods (i.e., A+, NLRTATV, and TLSR) are obviously worse than supervised-based approaches (i.e., SSPSR, ERCSR, Bi-3DQRNN, and DLGNN). On one hand, the advantage of traditional methods is can be applied to LR HSIs directly while deep-learning methods require lots of training samples, but the performance of traditional methods is limited by handcraft priors. In addition, deep-learning methods can achieve better reconstruction performance, but they heavily rely on training samples.

The proposed TSBSR method combines the strengths of both traditional and deep-learning methods. We obtain a pseudo-HR HSI by the pretrained band-wise super-resolution model and use an unsupervised scheme to estimate the spatial and spectral tensors in an end-to-end manner. In this way, our method can well generalize to a variety of degradations.

C. Experiments on Real Remotely Sensed HSI Data

We select the Pavia Centre and University dataset [47] to conduct the experiments on real remotely sensed HSI data. The methods of A+, NLRTATV, TLSR, and ours can be applied to the testing data directly, and the other competing methods (i.e., SSPSR, ERCSR, Bi-3DQRNN, and DLGNN) require training data. As for these four learning-based methods, we use Pavia Centre for training.

The testing data is from Pavia University, and we crop it to 128×128. The scale factor is 4. The results are obtained by applying different methods to the 128×128 Pavia University testing patch.

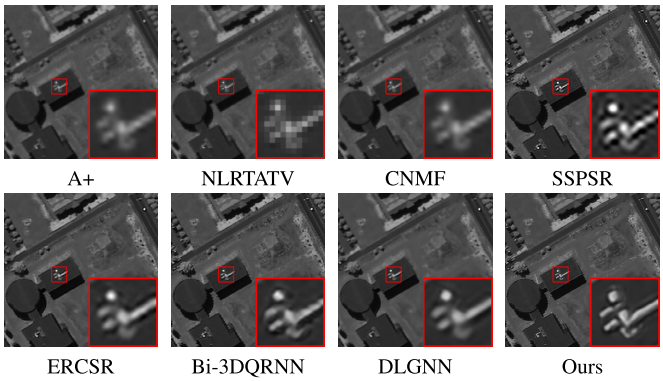


Fig. 7. HSI blind super-resolution results on Pavia University, and the scale factor is 4. We show the 100th band.

In Fig. 7, we show the reconstruction results. It can be observed that our method can recover more building textures than the competing methods. This indicates that our approach is more effective at reconstructing the fine details compared with the competing methods.

D. Ablation Study

The proposed method involves two steps, i.e., transfer learning and an unsupervised scheme. To demonstrate the effectiveness of both steps, we design ablation studies on transfer learning and unsupervised scheme. Without loss of generality, we evaluate our method with test degradation kernel $\mathbf{k}_1$ and Gaussian noise ($\sigma = 10$). The scale factor is fixed as 4, and the testing dataset is Pavia University [47].

1) Ablation Study on Transfer Learning: We introduce transfer learning for a decent initial result and overcome the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

TABLE VI

EFFECTIVENESS OF TRANSFER LEARNING ON THE PAVIA UNIVERSITY DATASET. WE USE TRAINING SET NAME (*Pavia* OR *AID*) TO REPRESENT OUR APPROACH WITHOUT THE TRANSFER LEARNING

Methods	PSNR ↑	SSIM ↑	SAM ↓
Bicubic	24.7444	0.7029	0.2259
SwinIR (Pavia)	24.7783	0.7150	0.2240
SwinIR (AID)	**25.0697**	**0.7575**	**0.1851**

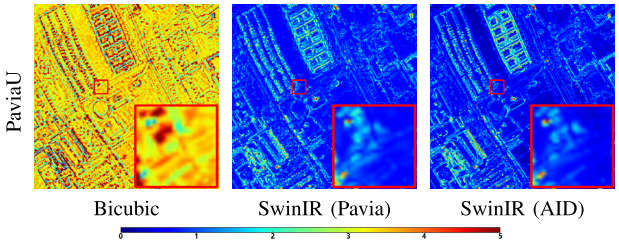


Fig. 8. Ablation study on transfer learning.

TABLE VII

EFFECTIVENESS OF UNSUPERVISED SCHEME AND DEGRADATION PRIOR. WE USE *NMF* TO REPRESENT THE NONNEGATIVE MATRIX NETWORK AND *D* DENOTES OUR DEGRADATION PRIOR. THE RESULT OF *Ours* IS EQUAL TO *SwinIR (AID)* + *NMF* + *D*

SwinIR (AID)	NMF	D	Metrics		
			PSNR ↑	SSIM ↑	SAM ↓
✓			25.0697	0.7575	0.1851
✓	✓		25.9421	0.7908	0.1242
✓	✓	✓	**26.1218**	**0.7944**	**0.1221**

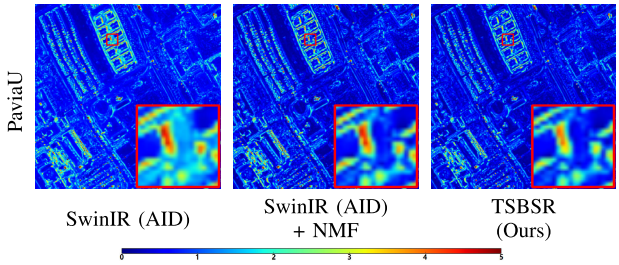


Fig. 9. Ablation study on the unsupervised scheme.

difficulties of acquiring a large amount of remotely sensed HSIs for training. Besides, there are more publicly available remotely sensed color image datasets with high-quality and quantity images compared to remotely sensed HSI datasets.

In the first step of the proposed method, we adopt a powerful single-image super-resolution model, i.e., SwinIR [44], to generate the pseudo-HR HSI. We choose the AID dataset [43] to train the SwinIR super-resolution model. The AID dataset contains 10 000 remotely sensed color images with the shape of $600 \times 600 \times 3$. It is customary to train the SwinIR model using the Pavia Centre dataset while the testing dataset is Pavia University. In the same way, to generate the training sets of both datasets, the number of training patches in the Pavia Centre dataset and the AID dataset are 707 and 1300 k, respectively.

In Table VI, we report the quantitative results of different approaches for training the SwinIR model to obtain the pseudo-HR HSI. We use *SwinIR (Pavia)* to represent the SwinIR model that is trained on the Pavia Centre dataset (707 training samples), and the *SwinIR (AID)* denotes the SwinIR model that is trained on the AID dataset (1300 k training samples).

We can see that training on the AID dataset outperforms training on Pavia Centre (i.e., +0.29 dB on the Pavia University dataset). The reconstruction results of the model trained at Pavia Centre are comparatively poorer than the model trained on AID. The visual results of Pavia University are shown in Fig. 8. We can see the clear margin of training on Pavia Centre and AID, the latter recovers more details of streets. This can be attributed to the limited data available in Pavia Centre, which demonstrates the effectiveness of the adopted transfer learning that could incorporate more training data.

2) Ablation Study on Unsupervised Scheme: However, direct use of the first step model, i.e., *SwinIR (AID)* trained by the transfer learning on the AID dataset, is suboptimal as it neglects the spectral correlations of different bands and the gap between predefined degradation and the real one. Therefore, in the second step, we propose an unsupervised scheme to further improve the results by enforcing the spectral correlations and narrowing the degradation gap.

The unsupervised scheme contains two components, including a nonnegative matrix factorization network and a degradation prior. Here, we evaluate their effectiveness. In Table VII, we use non-negative matrix factorization (*NMF*) to represent the nonnegative matrix factorization network and *D* to represent degradation prior. We can see that the *NMF* improves the performance by almost 1 dB on PSNR, and it also greatly reduces the SAM, which indicates that the spectral correlations are better preserved. The degradation prior also provides further improvement by considering the degradation gap.

Visual results are shown in Fig. 9. We can see that the "NMF" improves the performance by recovering more details of the building, and the degradation prior also provides further improvement.

V. CONCLUSION

In this article, we propose a TSBSR framework for blind super-resolution of single remotely sensed HSI, where the degradation from HR HSI to its LR version is unknown. The proposed method is suitable for data-insufficient scenarios and can generalize to various degradations without additional reference image input. We first train a single-channel super-resolution model on plenty of remotely sensed color image data, to generate a pseudo-HR HSI by transferring the spatial knowledge from the remotely sensed color image domain to the HSI domain. Then, to alleviate the spectral and spatial gaps between pseudo and latent HR HSIs, we build an unsupervised scheme to estimate the HR HSI and unknown degradation simultaneously. As for the spectral gap, we propose an end-to-end nonnegative matrix factorization network to estimate the endmember and abundance matrices of latent HR HSI. As for the spatial gap, we introduce a learnable degradation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

prior to modeling the process of degradation. Experiments on seven remotely sensed HSI datasets demonstrate the superior performance of the proposed method on single-blind HSI super-resolution. In the future, it is worth investigating deeper into this novel single HSI blind super-resolution task and the effectiveness of the pretrained super-resolution model in the whole framework.

REFERENCES

[1] X. Huang, M. Dong, J. Li, and X. Guo, "A 3-D-Swin transformer-based hierarchical contrastive learning method for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5411415.

[2] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[3] J. Liang, J. Zhou, X. Bai, and Y. Qian, "Salient object detection in hyperspectral imagery," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2013, pp. 2393–2397.

[4] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[5] M. Teke, H. S. Deveci, O. Haliloğlu, S. Z. Gürbüz, and U. Sakarya, "A short survey of hyperspectral remote sensing applications in agriculture," in *Proc. 6th Int. Conf. Recent Adv. Space Technol. (RAST)*, Jun. 2013, pp. 171–176.

[6] C. Wang et al., "A review of deep learning used in the hyperspectral image analysis for agriculture," *Artif. Intell. Rev.*, vol. 54, no. 7, pp. 5205–5253, Oct. 2021.

[7] Y. Cao and X. Huang, "A coarse-to-fine weakly supervised learning method for green plastic cover segmentation using high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 157–176, Jun. 2022.

[8] J. Li, X. Huang, and L. Tu, "WHU-OHS: A benchmark dataset for large-scale hersepctral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 113, Sep. 2022, Art. no. 103022.

[9] R. O. Green et al., "Imaging spectroscopy and the airborne visible/infrared imaging spectrometer (AVIRIS)," *Remote Sens. Environ.*, vol. 65, no. 3, pp. 227–248, Sep. 1998.

[10] J. M. Bioucas-Dias et al., "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[11] W. Huang, Z. Wu, H. Liu, L. Xiao, and Z. Wei, "Spatial–spectral compressive sensing for hyperspectral images super-resolution over learned dictionary," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 4930–4933.

[12] S. Tang, T. Xu, L. Huang, and L. Sun, "Hyperspectral image super-resolution via adaptive dictionary learning and double ℓ_1 constraint," *Remote Sens.*, vol. 11, no. 23, p. 2809, Nov. 2019.

[13] Y. Wang, X. Chen, Z. Han, and S. He, "Hyperspectral image super-resolution via nonlocal low-rank tensor approximation and total variation regularization," *Remote Sens.*, vol. 9, no. 12, p. 1286, Dec. 2017.

[14] J. Jiang, H. Sun, X. Liu, and J. Ma, "Learning spatial–spectral prior for super-resolution of hyperspectral imagery," *IEEE Trans. Comput. Imag.*, vol. 6, pp. 1082–1096, 2020.

[15] Y. Fu, Z. Liang, and S. You, "Bidirectional 3D quasi-recurrent neural network for hyperspectral image super-resolution," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2674–2688, 2021.

[16] Q. Li, Q. Wang, and X. Li, "Exploring the relationship between 2D/3D convolution for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8693–8703, Oct. 2021.

[17] Q. Wang, Q. Li, and X. Li, "Hyperspectral image superresolution using spectrum and feature context," *IEEE Trans. Ind. Electron.*, vol. 68, no. 11, pp. 11276–11285, Nov. 2021.

[18] Y. Fu, T. Zhang, L. Wang, and H. Huang, "Coded hyperspectral image reconstruction using deep external and internal learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 7, pp. 3404–3420, Jul. 2022.

[19] T. Zhang, Z. Liang, and Y. Fu, "Joint spatial–spectral pattern optimization and hyperspectral image reconstruction," *IEEE J. Sel. Topics Signal Process.*, vol. 16, no. 4, pp. 636–648, Jun. 2022.

[20] Z. Lai, K. Wei, and Y. Fu, "Deep plug-and-play prior for hyperspectral image restoration," *Neurocomputing*, vol. 481, pp. 281–293, Apr. 2022.

[21] K. Zhang, W. Zuo, and L. Zhang, "Deep plug-and-play super-resolution for arbitrary blur kernels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1671–1681.

[22] S. Anwar, N. Barnes, and L. Petersson, "Attention-based real image restoration," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Dec. 13, 2021, doi: 10.1109/TNNLS.2021.3131739.

[23] W. Wang, W. Zeng, Y. Huang, X. Ding, and J. Paisley, "Deep blind hyperspectral image fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4149–4158.

[24] L. Zhang, J. Nie, W. Wei, Y. Li, and Y. Zhang, "Deep blind hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2388–2400, Jun. 2021.

[25] J. Liu, Z. Wu, L. Xiao, and X.-J. Wu, "Model inspired autoencoder for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522412.

[26] Y. Yuan, X. Zheng, and X. Lu, "Hyperspectral image superresolution by transfer learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 5, pp. 1963–1974, May 2017.

[27] Z. He and L. Liu, "Hyperspectral image super-resolution inspired by deep Laplacian pyramid network," *Remote Sens.*, vol. 10, no. 12, p. 1939, Dec. 2018.

[28] C. Zou and X. Huang, "Hyperspectral image super-resolution combining with deep learning and spectral unmixing," *Signal Process., Image Commun.*, vol. 84, May 2020, Art. no. 115833.

[29] J. Hu, Y. Li, and W. Xie, "Hyperspectral image super-resolution by spectral difference learning and spatial error correction," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1825–1829, Oct. 2017.

[30] W. Xie, X. Jia, Y. Li, and J. Lei, "Hyperspectral image super-resolution using deep feature matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6055–6067, Aug. 2019.

[31] S. Mei, X. Yuan, J. Ji, Y. Zhang, S. Wan, and Q. Du, "Hyperspectral image spatial super-resolution via 3D full convolutional neural network," *Remote Sens.*, vol. 9, no. 11, p. 1139, Nov. 2017.

[32] Y. Qu, H. Qi, and C. Kwan, "Unsupervised sparse Dirichlet-Net for hyperspectral image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2511–2520.

[33] Y. Fu, T. Zhang, Y. Zheng, D. Zhang, and H. Huang, "Hyperspectral image super-resolution with optimized RGB guidance," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11653–11662.

[34] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2016, pp. 391–407.

[35] X. Lu, D. Yang, J. Zhang, and F. Jia, "Hyperspectral image super-resolution based on spatial correlation-regularized unmixing convolutional neural network," *Remote Sens.*, vol. 13, no. 20, p. 4074, Oct. 2021.

[36] R. Keys, "Cubic convolution interpolation for digital image processing," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-29, no. 6, pp. 1153–1160, Dec. 1981.

[37] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[38] T. Michaeli and M. Irani, "Nonparametric blind super-resolution," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 945–952.

[39] J. Gu, H. Lu, W. Zuo, and C. Dong, "Blind super-resolution with iterative kernel correction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1604–1613.

[40] Z. Luo, Y. Huang, S. Li, L. Wang, and T. Tan, "Unfolding the alternating optimization for blind super resolution," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 5632–5643.

[41] X. Kang, J. Li, P. Duan, F. Ma, and S. Li, "Multilayer degradation representation-guided blind super-resolution for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5534612.

[42] Z. Liu, R. Feng, L. Wang, W. Han, and T. Zeng, "Dual learning-based graph neural network for remote sensing image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628614.

[43] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[44] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.

[45] K. Wei, Y. Fu, and H. Huang, "3-D quasi-recurrent neural network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 363–375, Jan. 2021.

[46] M. Arngren, M. N. Schmidt, and J. Larsen, "Unmixing of hyperspectral images using Bayesian non-negative matrix factorization with volume prior," *J. Signal Process. Syst.*, vol. 65, no. 3, pp. 479–496, Dec. 2011.

[47] P. Gamba, "A collection of data for urban area characterization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2004, p. 72.

[48] F. Pacifici, Q. Du, and S. Prasad, "Report on the 2013 IEEE GRSS data fusion contest: Fusion of hyperspectral and LiDAR data [Technical Committees]," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 3, pp. 36–38, Sep. 2013.

[49] Y. Xu et al., "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 IEEE GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1709–1724, Jun. 2019.

[50] R. Timofte, V. De Smet, and L. Van Gool, "A+: Adjusted anchored neighborhood regression for fast super-resolution," in *Proc. Asian Conf. Comput. Vis.* Cham, Switzerland: Springer, Jan. 2015, pp. 111–126.

[51] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[52] R. H. Yuhas, J. W. Boardman, and A. F. Goetz, "Determination of semi-arid landscape endmembers and seasonal trends using convex geometry spectral unmixing techniques," in *Proc. Summaries 4th Annu. JPL Airborne Geosci. Workshop*, Oct. 1993, pp. 1–6.

[53] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, 2019, pp. 1–19.



Zhiyuan Liang received the bachelor's degree from the Hefei University of Technology, Hefei, China, in 2020, and the M.S. degree from the Beijing Institute of Technology, Beijing, China, in 2023.

Her research interests lie primarily in the area of hyperspectral image processing, computer vision, and deep learning.



Shuai Wang received the B.E. degree from Jilin University, Changchun, China, in 2007, and the Ph.D. degree from the School of Instrumentation and Optoelectronic Engineering, Beihang University, Beijing, China, in 2012.

She is currently an Associate Professor with the School of Computer Science and Engineering, Beihang University. Her research interests include computer vision, intelligent perception, and AIoT.



Tao Zhang (Student Member, IEEE) received the B.S. and Ph.D. degrees from the School of Computer Science and Technology, Beijing Institute of Technology, Beijing, China, in 2017 and 2023, respectively.

His research interests include deep learning, image processing, and computational photography.



Ying Fu (Senior Member, IEEE) received the B.S. degree in electronic engineering from Xidian University, Xi'an, China, in 2009, the M.S. degree in automation from Tsinghua University, Beijing, China, in 2012, and the Ph.D. degree in information science and technology from The University of Tokyo, Tokyo, Japan, in 2015.

She is currently a Professor with the School of Computer Science and Technology, Beijing Institute of Technology. Her research interests include remote sensing image processing, computer vision, and computational photography.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:37:36 UTC from IEEE Xplore. Restrictions apply.